

Лабораторная работа №1

Имитационное моделирование

бахи сиди али темассини

2026-09-30

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Раздел 1

Информация

- бахи сиди али темассини
- Российский университет дружбы народов

Раздел 2

Цель работы

Цель работы

- Подготовить рабочее окружение для имитационного моделирования.
- Освоить организацию проекта на языке Julia.
- Реализовать модель экспоненциального роста.
- Провести вычислительные эксперименты для различных значений параметра роста.
- Проанализировать полученные результаты.

Раздел 3

Настройка Git и средств аутентификации

Базовая настройка Git

- Проверена установка Git.
- Установлен GitHub CLI.
- Настроены глобальные параметры Git.

```
sidiali@fedora:~$ sudo dnf -y install git
[sudo] password for sidiali:
Updating and loading repositories:
  Fedora 44 - x86_64 - Updates
  RPM Fusion for Fedora 44 - Nonfree - Steam
  RPM Fusion for Fedora 44 - Nonfree - NVIDIA Driver
  google-chrome
  Copr repo for PyCharm owned by phracek
  Copr repo for rstudio owned by iucar
Repositories loaded.
Package "git-2.55.0-1.fc44.x86_64" is already installed.

Nothing to do.
sidiali@fedora:~$ ^[[200~sudo dnf -y install gh
bash: sudo: command not found...
sidiali@fedora:~$ ~sudo dnf -y install gh
bash: ~sudo: command not found...
Similar command is: 'sudo'
sidiali@fedora:~$ sudo dnf -y install gh
Updating and loading repositories:
Repositories loaded.
Package "gh-2.97.0-2.fc44.x86_64" is already installed.
```

Аутентификация на GitHub

- Выполнена аутентификация через GitHub CLI.
- Настроен протокол HTTPS для операций Git.

```
sidiali@fedora:~$ gh auth login
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: 9647-6847
Press Enter to open https://github.com/login/device in your browser...
restorecon: SELinux: Could not get canonical path for /home/sidiali/.config/mozilla/
Safe Browsing server returned a 400 during update:request url = https://safebrowsing
sVuuco&$httpMethod=POST, payload = ChUKE25hdmNsaWVudC1hdXRvLWZmb3gaCggFEALiAiACKAEaC
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as sidiali2030
sidiali@fedora:~$
```

Рисунок 2: Успешная аутентификация на GitHub через GitHub CLI

Создание SSH-ключа

- Проверено состояние авторизации командой `gh auth status`.
- Создана пара SSH-ключей Ed25519.

```
sidiali@fedora:~$ gh auth status
github.com
✓ Logged in to github.com account sidiali2030 (keyring)
- Active account: true
- Git operations protocol: https
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo', 'workflow'
sidiali@fedora:~$ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519-git -C "sidialitemassinibahi@gmail.com"
Generating public/private ed25519 key pair.
Enter passphrase for "/home/sidiali/.ssh/id_ed25519-git" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/sidiali/.ssh/id_ed25519-git
Your public key has been saved in /home/sidiali/.ssh/id_ed25519-git.pub
The key fingerprint is:
SHA256:AGZ0QPw4oyavWhdubT60uzFvp+bD2pEhL7EcD7L8YQ8 sidialitemassinibahi@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|   +B..          |
|   o.o           |
|    o.           |
|   + ..         |
|  ..*o. S       |
|..o= @ o        |
| ++ E.B         |
| ..= &*..        |
|+.  *00=        |
+----[SHA256]-----+
sidiali@fedora:~$
```

Создание GPG-ключа

- Создан GPG-ключ с данными пользователя.
- Созданы открытый и секретный ключи.

```
Real name: sidiali
Email address: sidialitemassinibahi@gmail.com
Comment: key GPG
You selected this USER-ID:
    "sidiali (key GPG) <sidialitemassinibahi@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? 0
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/sidiali/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/sidiali/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/sidiali/.gnupg/openpgp-revocs.d/0021A3E678288A653044C2F44A
public and secret key created and signed.

pub   rsa4096 2026-09-29 [SC]
      0021A3E678288A653044C2F44A009A20BCB87699
uid           sidiali (key GPG) <sidialitemassinibahi@gmail.com>
sub   rsa4096 2026-09-29 [E]

sidiali@fedora:~$
```

Настройка GPG-подписания

- Проверен созданный GPG-ключ.
- Настроено автоматическое GPG-подписание коммитов.

```
sidiali@fedora:~$ gpg --armor --export 0021A3E678288A653044C2F44A009A20BCB87699 | wl-copy
sidiali@fedora:~$ gpg --armor --export 4A009A20BCB87699 | wl-copy
sidiali@fedora:~$ gpg --armor --export 0021A3E678288A653044C2F44A009A20BCB87699 | wl-copy
sidiali@fedora:~$ git config --global user.signingkey 0021A3E678288A653044C2F44A009A20BCB87699
sidiali@fedora:~$ git config --global commit.gpgsign true
sidiali@fedora:~$ git config --global gpg.program $(which gpg)
sidiali@fedora:~$
```

Рисунок 5: Настройка GPG-подписания коммитов Git

Проверка инструментов

- Проверены установленные инструменты рабочего окружения.
- Подтверждено наличие `commitizen`, `cz-customizable` и `standard-version`.

```
Done in 19.7s using pnpm v10.33.0
sidiali@fedora:~$ yarn global add standard-version
bash: yarn: command not found...
Packages providing this file are:
'nodejs-corepack'
'yarnpkg'
sidiali@fedora:~$ quarto --version
1.10.18
sidiali@fedora:~$
```

Рисунок 6: Проверка версий и установленных инструментов рабочего окружения

Раздел 4

Подготовка рабочего пространства лабораторной работы

Клонирование репозитория

- Репозиторий курса клонирован с GitVerse по SSH.
- Создана локальная копия проекта.

```
and the repository exists.  
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling$ git clone ssh://git@gitverse.ru:2222/sidiali2030/2026-1--study--simulation-modeling.git  
Cloning into '2026-1--study--simulation-modeling'...  
remote: Enumerating objects: 42, done.  
remote: Counting objects: 100% (42/42), done.  
remote: Compressing objects: 100% (41/41), done.  
remote: Total 42 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (42/42), 27.57 KiB | 130.00 KiB/s, done.  
Resolving deltas: 100% (1/1), done.  
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling$
```

Рисунок 7: Клонирование репозитория курса с GitVerse

Подготовка структуры проекта

- Выполнена команда `make prepare`.
- Загружены подмодули шаблонов презентации и отчёта.
- Проверена структура лабораторных работ.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ make prepare
Submodule 'template/presentation' (https://gitverse.ru/dharma/academic-presentation-markdown-template.git) registered for path 'template/presentation'
Submodule 'template/report' (https://gitverse.ru/dharma/academic-laboratory-report-template.git) registered for path 'template/report'
Cloning into '/home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/template/presentation'...
Cloning into '/home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/template/report'...
Submodule path 'template/presentation': checked out '160e4f43014e4b011f431eb0ea1c10f98bdd6ebe'
Submodule path 'template/report': checked out '3818a93e0fd31868c9a5c0b2747ce9370e0ec1a8'
Synchronizing submodule url for 'template/presentation'
Synchronizing submodule url for 'template/report'
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ ls
CODE_OF_CONDUCT.md  CONTRIBUTING.md  COURSE  labs  LICENSE  Makefile  package.json  prepare  README.en.md  README.git-flow.md  README.md  SECURITY.md  template
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ ls labs/
lab01  lab02  lab03  lab04  lab05  lab06  lab07  lab08  README.md  README.ru.md
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$
```

Рисунок 8: Подготовка проекта и проверка структуры лабораторных работ

Синхронизация с GitVerse

- Изменения отправлены в GitVerse.
- Ветка `master` синхронизирована с `origin/master`.
- Рабочее дерево не содержит изменений.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push
Enumerating objects: 72, done.
Counting objects: 100% (72/72), done.
Delta compression using up to 6 threads
Compressing objects: 100% (55/55), done.
Writing objects: 100% (68/68), 839.23 KiB | 32.28 MiB/s, done.
Total 68 (delta 15), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To ssh://gitverse.ru:2222/sidiali2030/2026-1--study--simulation-modeling.git
   ff9aab6..73615c5  master -> master
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$
```

Рисунок 9: Отправка изменений в GitVerse и проверка состояния репозитория

- Ветка `master` отправлена в репозиторий GitHub.

```
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling$ git push github master
Enumerating objects: 110, done.
Counting objects: 100% (110/110), done.
Delta compression using up to 6 threads
Compressing objects: 100% (95/95), done.
Writing objects: 100% (110/110), 866.70 KiB | 2.79 MiB/s, done.
Total 110 (delta 16), reused 41 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (16/16), done.
To github.com:sidiali2030/2026-1--study--simulation-modeling.git
 * [new branch]      master -> master
sidiali@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling$
```

Рисунок 10: Отправка ветки `master` в удалённый репозиторий GitHub

Инициализация Git Flow

- Инициализирован Git Flow.
- Основная ветка — master.
- Ветка разработки — develop.
- Префикс тегов версий — v.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/.git/hooks]
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$
```

Рисунок 11: Инициализация и настройка Git Flow

Создание релиза

- Ветки и тег v1.0.0 отправлены в GitVerse и GitHub.
- Сформирован CHANGELOG.md.
- Создан релиз v1.0.0.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: You can create a PR using the link below:
remote:   https://gitverse.ru/sidiali2030/2026-1--study--simulation-modeling/compare?baseBranch=master&headBranch=release/1.0.0
remote:
remote: . Processing 1 references
remote: Processed 1 references in total
To ssh://gitverse.ru:2222/sidiali2030/2026-1--study--simulation-modeling.git
 * [new branch]   release/1.0.0 -> release/1.0.0
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ ^[[200~git push --all
bash: git: command not found...
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push --tags
Everything up-to-date
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push github --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sidiali2030/2026-1--study--simulation-modeling.git
 * [new branch]   develop -> develop
 * [new branch]   release/1.0.0 -> release/1.0.0
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling$ git push github --tags
Everything up-to-date
```

Рисунок 12: Публикация веток и тега v1.0.0 и создание релиза

Раздел 5

Создание и проверка проекта моделирования

Создание проекта Julia

- Создан каталог `project`.
- Проверена структура проекта.
- Созданы каталоги `data`, `notebooks`, `plots`, `scripts`, `src` и `test`.

```
[0e850ede] + nghttp2_jll v1.64.0+1
[3f19e933] + p7zip_jll v17.5.0+2
Проект создан: project
Перейдите в директорию: cd project
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$ ls
presentation project report setup_project.jl
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$ ls pr
presentation/ project/
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$ ls project/
data Manifest.toml notebooks papers plots Project.toml README.md _research scripts src test
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01$
```

Рисунок 13: Создание и проверка структуры проекта Julia

Установка зависимостей

- Установлены зависимости Julia.
- Выполнена предварительная компиляция пакетов.

```
[bea87d4a] + SuiteSparse_jll v7.8.3+2
[be850b98] + libblastrampoline_jll v5.15.0+0
Info Packages marked with ⚠ and ⚠ have new versions available. Those with ⚠ may be upgradable, but those with ⚠ are restricted by compatibility constraints from upgrading. To
use `status --outdated -m`
Building Conda => `./julia/scratchspaces/44cfe95a-1eb2-52ea-b672-e2afcf69b78f/af86b0cfa4c514c7b9546756dbae91fcfbc92dc9/build.log`
Building IJulia => `./julia/scratchspaces/44cfe95a-1eb2-52ea-b672-e2afcf69b78f/102656c4efc9737f892e1bca7e66ae374c658740/build.log`
Precompiling packages finished.
291 dependencies successfully precompiled in 622 seconds. 58 already precompiled.
2 dependencies had output during precompilation:
LinearSolve
  Downloading artifact: MKL
MKL_jll
  Downloading artifact: IntelOpenMP
  Downloading artifact: oneTBB

Все пакеты установлены!
Для проверки: using DrWatson, DifferentialEquations, Plots
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$
```

Рисунок 14: Установка и предварительная компиляция пакетов Julia

Проверка окружения

- Выполнен `scripts/test_setup.jl`.
- Проверена доступность необходимых пакетов.
- Проверена структура каталогов проекта.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/test_setup.jl
[✓] Проект активирован: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project

Проверка пакетов:
✓ DrWatson
✓ DifferentialEquations
✓ Plots
✓ DataFrames
✓ CSV
✓ JLO2
✓ Literate
✓ IJulia
✓ BenchmarkTools
✓ Quarto

Структура проекта:
Корень: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project
Данные: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/data
Скрипты: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/src
Графики: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/plots
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$
```

Рисунок 15: Проверка окружения и установленных пакетов проекта

Раздел 6

Реализация модели экспоненциального роста

- Создан сценарий `01_exponential_growth.jl`.
- Получены первые значения решения модели.
- Рассчитано время удвоения: `2.31`.

```
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ cd scripts/
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts$ nano 01_exponential_growth.jl
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts$ cd ..
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/01_exponential_growth.jl
Could not create decoration from factory! Running with no decorations.
Первые 5 строк результатов:
5x2 DataFrame
  Row  t      u
     Float64  Float64
1     0.0  1.0
2     0.1  1.03045
3     0.2  1.06184
4     0.3  1.09417
5     0.4  1.1275

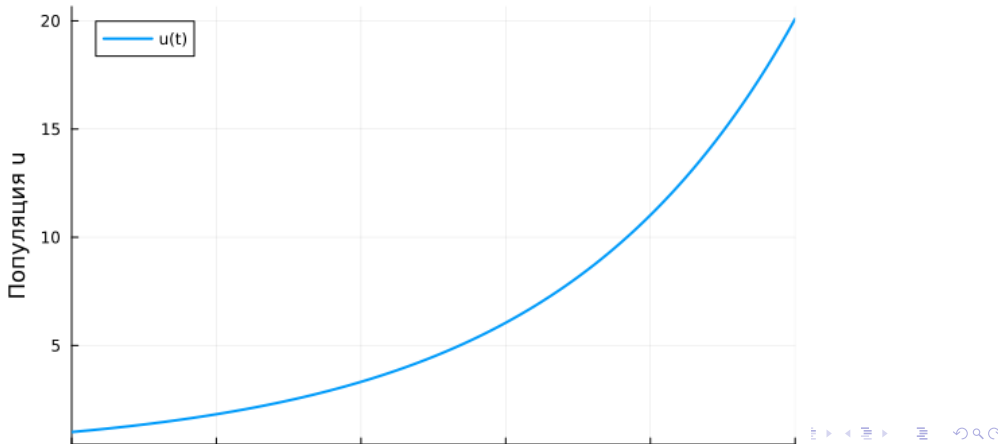
Аналитическое время удвоения: 2.31
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls
add_packages.jl  data  Manifest.toml  notebooks  plots  Project.toml  README.md  _research  scripts  src  test
sidial@fedora:~/work/study/2026-1/2026-1--study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls plots/
'exponential_growth_u0.3.png'
```

Рисунок 16: Выполнение модели экспоненциального роста

Визуализация модели

- Построен график для $\alpha = 0.3$.
- Получен экспоненциальный рост популяции во времени.

Экспоненциальный рост ($\alpha = 0.3$)



Сохранение результатов

- Результаты сохранены в `all_results.jld2`.
- График сохранён в каталоге результатов.

```
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$ nano scripts/01_exponential_growth.jl
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$ julia --project=. scripts/01_exponential_growth.jl
Could not create decoration from factory! Running with no decorations.
Переве 5 строк результатов:
5x2 DataFrame
  Row  t      u
  ----  -
1     0.0  1.0
2     0.1  1.83845
3     0.2  1.06184
4     0.3  1.09417
5     0.4  1.1275

Аналитическое время удвоения: 2.31
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$ ls plots/
01_exponential_growth_ "exponential_growth_u=0.3.png"
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$ ls data/01_exponential_growth/
all_results.jld2
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$ ls plots/01_exponential_growth/
'exponential_growth_u=0.3.png'
sidiall@fedora:~/work/study/2026-1/2026-1==study==simulation-modeling/2026-1==study==simulation-modeling/labs/lab01/project$
```

Рисунок 18: Проверка сохранённых данных и графика модели

Раздел 7

Литературная реализация модели

Генерация файлов

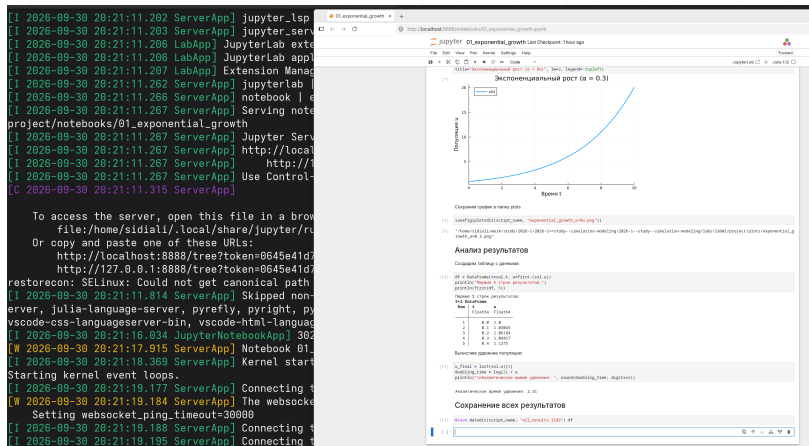
- Подготовлен сценарий `tangle.jl`.
- Сформирован чистый код Julia.
- Созданы документы `.qmd` и `.ipynb`.

```
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ nano scripts/tangle.jl
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls scripts/
01_exponential_growth.jl  01_exponential_growth.ly  intro.jl  tangle.jl  test_setup.jl
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/tangle.jl scripts/01_exponential_
jl
Генерация из: scripts/01_exponential_growth.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth/01_exponential_gr
jl
√ Чистый скрипт: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth/01_exponential_
jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/markdown/01_exponential_growth/01_exponential_gr
qmd
√ Quarto: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/markdown/01_exponential_growth/01_exponential_growth
ipynb
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/01_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/notebooks/01_exponential_growth/01_exponential_g
pynb"
√ Notebook: /home/sidiali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/notebooks/01_exponential_growth/01_exponential_gr
nb
Готово! Все файлы созданы.
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls notebooks/
01_exponential_growth
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls notebooks/01_exponential_growth/
01_exponential_growth.ipynb
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ ls markdown/01_exponential_growth/
01_exponential_growth.qmd
sidiali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$
```

Рисунок 19: Генерация кода, документации Quarto и Jupyter Notebook

Выполнение Jupyter Notebook

- Выполнен 01_exponential_growth.ipynb.
- Получены график и таблица результатов.
- Получено время удвоения 2.31 .



Раздел 8

Проведение вычислительного эксперимента

Параметрическое сканирование

- Выполнен 02_exponential_growth.jl.
- Базовый эксперимент проведён при $\alpha = 0.3$.
- Исследованы $\alpha = 0.1, 0.3, 0.5, 0.8, 1.0$.
- Результаты сведены в таблицу.

```
sidali@fedora:~/work/study/2026-1==study--simulation-modeling/2026-1==study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/02_exponential_growth.jl
Базовые параметры эксперимента:
α = 0.3
u0 = [1.0]
saveat = 0.1
solver = Tsit5{typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!), FastBroadcast.Serial}(OrdinaryDiffEqCore.trivial_limiter!, OrdinaryDiffEqCore.trivial_limiter!, FastBroadcast.Serial())
experiment_name = base_experiment
tspan = (0.0, 10.0)
[ Info: File /home/sidali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1==study--simulation-modeling/labs/lab01/project/data/02_exponential_growth/single/exp-growth_experiment.n
ame=base_experiment_saveat=0.1_α=0.3.jld2 does not exist. Producing it now...
[ Info: File /home/sidali/work/study/2026-1/2026-1==study--simulation-modeling/labs/lab01/project/data/02_exponential_growth/single/exp-growth_experiment.n
ame=base_experiment_saveat=0.1_α=0.3.jld2 saved.

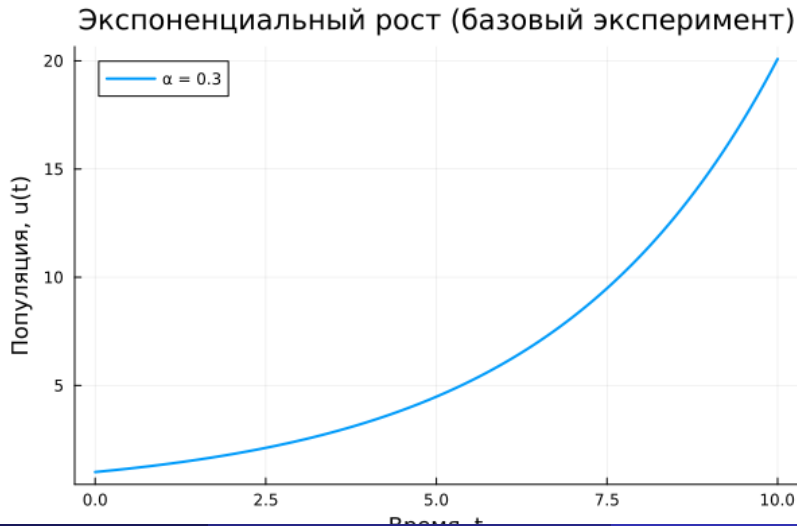
Результаты базового эксперимента:
Финальная популяция: 28.089448516186734
Время удвоения: 2.31
Файл результатов: /home/sidali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1==study--simulation-modeling/labs/lab01/project/data/02_exponential_growth/single/exp-growth_experi
ment_name=base_experiment_saveat=0.1_α=0.3.jld2
Could not create decoration from factory! Running with no decorations.

=====
ПАРАМЕТРИЧЕСКОЕ СКАНИРОВАНИЕ
Всего комбинаций параметров: 5
Исследуемые значения α: [0.1, 0.3, 0.5, 0.8, 1.0]
=====
r Warning: Assignment to `data` in soft scope is ambiguous because a global variable by the same name exists: `data` will be treated as a new local. Disambiguate by using `local data` to supp
ress this warning or `global data` to assign to the existing global variable.
julia -i ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1==study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl:109
r Warning: Assignment to `path` in soft scope is ambiguous because a global variable by the same name exists: `path` will be treated as a new local. Disambiguate by using `local path` to supp
ress this warning or `global path` to assign to the existing global variable.
julia -i ~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1==study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl:109
Прогресс: 1/5 | α = 0.1
Прогресс: 2/5 | α = 0.3
Прогресс: 3/5 | α = 0.5
Прогресс: 4/5 | α = 0.8
Прогресс: 5/5 | α = 1.0
```

Рисунок 21: Выполнение базового эксперимента и параметрического сканирования модели

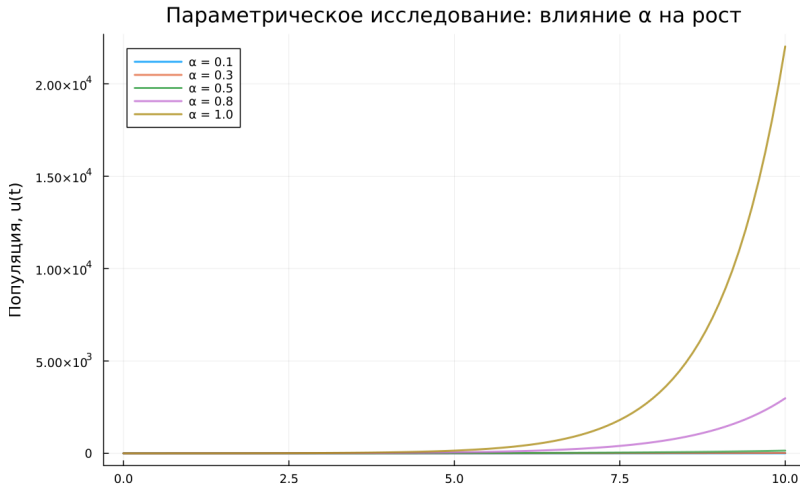
Базовый эксперимент

- Построен график базового эксперимента при $\alpha = 0.3$.



Сравнение параметров модели

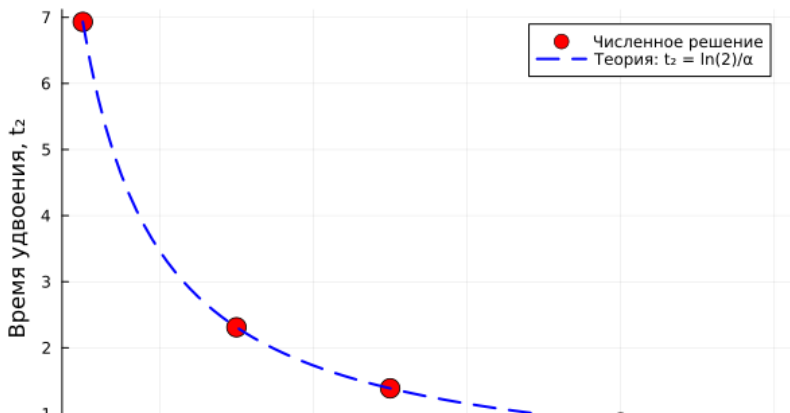
- Выполнено сравнение результатов для различных α .
- Увеличение α приводит к более быстрому росту популяции.



Анализ времени удвоения

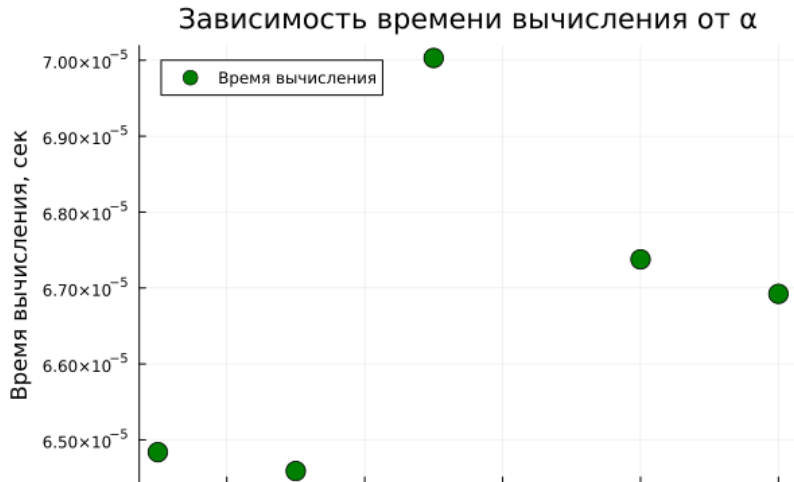
- Построена зависимость времени удвоения от α .
- Численные результаты сравнены с $t_2 = \ln(2)/\alpha$.
- При увеличении α время удвоения уменьшается.

Зависимость времени удвоения от α



Анализ времени вычисления

- Построена зависимость времени вычисления от α .
- Сравнено время вычислений для пяти значений параметра.



Раздел 9

Литературная реализация вычислительного эксперимента

Генерация файлов эксперимента

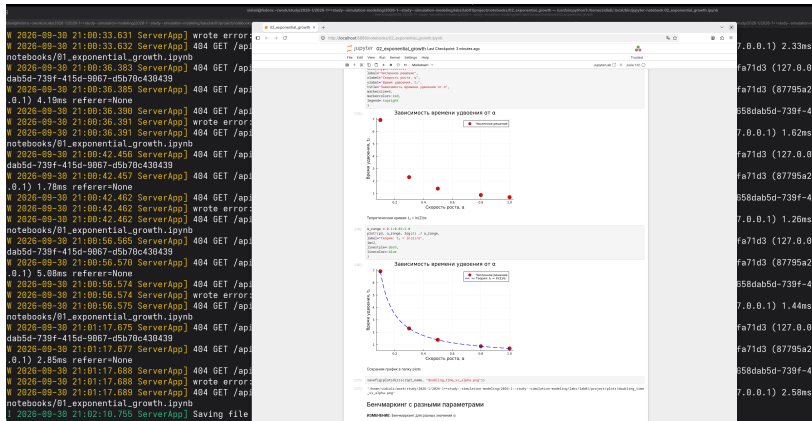
- 02_exponential_growth.jl преобразован с помощью tangle.jl.
- Создан чистый сценарий Julia.
- Созданы документ Quarto и Jupyter Notebook.

```
sidali@fedora:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project$ julia --project=. scripts/tangle.jl scripts/02_exponential_growth.jl
Генерация из: scripts/02_exponential_growth.jl
[ Info: generating plain script file from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth/02_exponential_growth.jl"
✓ Чистый скрипт: /home/sidali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth/02_exponential_growth.jl
[ Info: generating markdown page from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/markdown/02_exponential_growth/02_exponential_growth.qmd"
✓ Quarto: /home/sidali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/markdown/02_exponential_growth/02_exponential_growth.qmd
[ Info: generating notebook from "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/scripts/02_exponential_growth.jl"
[ Info: writing result to "~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/notebooks/02_exponential_growth/02_exponential_growth.ipynb"
✓ Notebook: /home/sidali/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab01/project/notebooks/02_exponential_growth/02_exponential_growth.ipynb
Готово! Все файлы созданы.
```

Рисунок 26: Генерация файлов литературной реализации вычислительного эксперимента

Выполнение параметрического исследования

- Выполнен 02_exponential_growth.ipynb.
- Получены результаты параметрического исследования.
- Выполнено сравнение численного и теоретического времени удвоения.



Раздел 10

Выводы

- Подготовлено рабочее окружение.
- Настроены Git, GitHub и GitVerse.
- Создан и проверен проект Julia.
- Реализована модель экспоненциального роста.
- Проведено параметрическое исследование.
- При увеличении λ рост популяции ускоряется.
- При увеличении λ время удвоения уменьшается.
- Сформированы сценарии, документация Quarto и Jupyter Notebook.